

Problem Statement & Stakeholder Analysis

E-commerce & Campaign Profitability Analyzer

Business Problem

The project addresses a disconnect between Performance Marketing and Corporate Finance reporting.

Marketing teams evaluate campaign performance using top-of-funnel or gross checkout measures, while Finance requires a reconciled view of actual economics after accounting for:

- Discounts
- Product costs (COGS)
- Cancellations
- Refunds
- Payments
- Marketing spend

This difference results in inconsistent views of revenue, profitability, cash realization, and marketing performance across teams.

Proposed Solution

The Marketing & Financial Unit Economics Engine creates a requirements-driven reporting and analytics layer connecting:

Marketing → Customers → Orders → Products → Payments → Refunds → Financial Analysis

The system uses a relational database and analytical SQL to calculate consistent business metrics, including:

- Net Revenue
- Cost of Goods Sold (COGS)
- Contribution Profit
- Return on Ad Spend (ROAS)
- Customer Lifetime Spend
- Customer Dormancy
- Month-over-Month Revenue Growth

Business Objective

The primary objective is to provide Marketing, Finance, Product, and CRM stakeholders with a common and reconciled view of business performance.

The project demonstrates the complete Business Analyst lifecycle:

Business Problem → Requirements → Data Model → Business Rules → SQL → Testing → Reporting → Business Insights

Key Business Questions

The analysis answers the following six questions, each directly implemented and validated in SQL:

1. Which marketing platforms generate the highest contribution profit?
2. What is the reconciled ROAS by platform?
3. Which customers have lifetime qualifying spend of at least \$1,500 (VIP)?
4. Which VIP customers have been dormant for 60+ days?
5. What is monthly revenue and its Month-over-Month growth?
6. Which orders have payment or refund discrepancies requiring reconciliation?

Scope note: an earlier draft of this document listed additional questions covering refund rate by product/category, payment-method-level reconciliation, spend-vs-contribution campaign flags, and new/returning/dormant segmentation. These were scoped out of the current build to keep the analysis focused and fully validated end-to-end, rather than partially covering a longer list. They are logged below as candidate extensions.

Scoped-Out Extensions (Not Built in This Version)

- *Refund rate by product or category*
- *Payment method with the largest unreconciled amount*
- *Campaigns with high spend but weak contribution, as an explicit flag beyond ranking*
- *Customer segmentation as New / Returning / Dormant (current build covers VIP / Regular × Active / Dormant / Never Ordered only)*

Stakeholder Overview

Stakeholder	Role / Interest	Key Information Needed
Finance Team	Monitor actual financial performance and profitability	Net revenue, COGS, refunds, payments, contribution profit
Marketing Team	Evaluate campaign and channel performance	Ad spend, revenue, ROAS, contribution profit
Marketing Leadership	Make campaign and channel allocation decisions	Channel profitability, campaign performance, financial contribution
Product Team	Understand product-level economics	Product revenue, discounts, COGS, contribution margin
CRM / Retention Team	Identify valuable and dormant customers	Customer lifetime spend, tier, last order, dormancy
Business Analyst	Gather requirements and translate business needs into the analytical solution	Requirements, business rules, data model, SQL logic, testing
Management / Executives	Monitor overall business performance	Revenue growth, profitability, campaign performance, customer and product insights

Stakeholder Requirements

1. Finance Team

Finance requires a reconciled view of business economics after considering discounts, COGS, cancellations, refunds, payments, and marketing spend.

Key questions:

- What is the actual net revenue?
- What is the contribution profit?
- What amount was refunded?
- Are payments reconciled with orders?

2. Marketing Team

Marketing requires visibility into the financial performance of campaigns and advertising platforms.

Key questions:

- Which platforms generate the most contribution?
- What is the ROAS by platform?
- How does campaign performance rank across platforms?

3. CRM / Retention Team

The CRM team needs customer-level information to identify valuable and potentially dormant customers.

Key questions:

- Who are the highest-value customers?
- Which customers qualify as VIP?
- Which VIP customers have been dormant for 60+ days?

4. Management / Executives

Management requires a consolidated view of business performance for reporting and decision-making.

Key questions:

- How is revenue changing month over month?
- Which areas of the business require further investigation?

Business Analyst Responsibility

The Business Analyst acts as the link between business stakeholders and the technical implementation.

Responsibilities include:

- Gathering and documenting stakeholder requirements
- Converting business needs into functional requirements
- Defining business rules and metric definitions
- Coordinating the relational data model
- Translating requirements into analytical SQL use cases
- Defining testing scenarios and expected results
- Maintaining traceability from requirements through implementation and testing
- Supporting final reporting and business insights

Stakeholder-to-Requirement Mapping

Stakeholder	Primary Requirements
Finance	Profitability, COGS, payments, refunds, reconciliation
Marketing	Campaign profitability, ROAS, channel ranking
CRM / Retention	Customer tiers and dormancy
Management	Revenue growth and executive reporting
Business Analyst	Requirements, business rules, testing and traceability